

Boolean Algebra

Logic to gates

The big three

- AND
- OR
- NOT
- Special mention: XOR

Truth tables - AND

A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1

Truth tables - OR

A	B	Q
0	0	0
0	1	1
1	0	1
1	1	1

Truth tables - NOT

A	Q (A')
0	1
1	0

Truth tables - XOR

A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0

Next level!

- We can combine multiple statements and therefore multiple truth tables!
- $A \text{ AND } B \text{ OR } C$?
- $(A \text{ AND } B) \text{ or } C$ (assumed)
- $A \text{ AND } (B \text{ OR } C)$ (use parentheses)

Convention

- Like algebra – “AND” is similar to “multiplication” and has higher “precedence”
- $A \text{ AND } B \text{ OR } C$
- $A.B + C$

Example

- Swipe cards and access privileges are a good real world example.
- My card will open this room IF it is during semester AND I am a staff member or student

Assign variables

- $A = \text{during semester}$
- $B = \text{staff}$
- $C = \text{student}$
- $A \text{ AND } (B \text{ OR } C)$

Full table!

A	B	C	Q
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

(B OR C) first

B	C	R
0	0	0
0	1	1
1	0	1
1	1	1

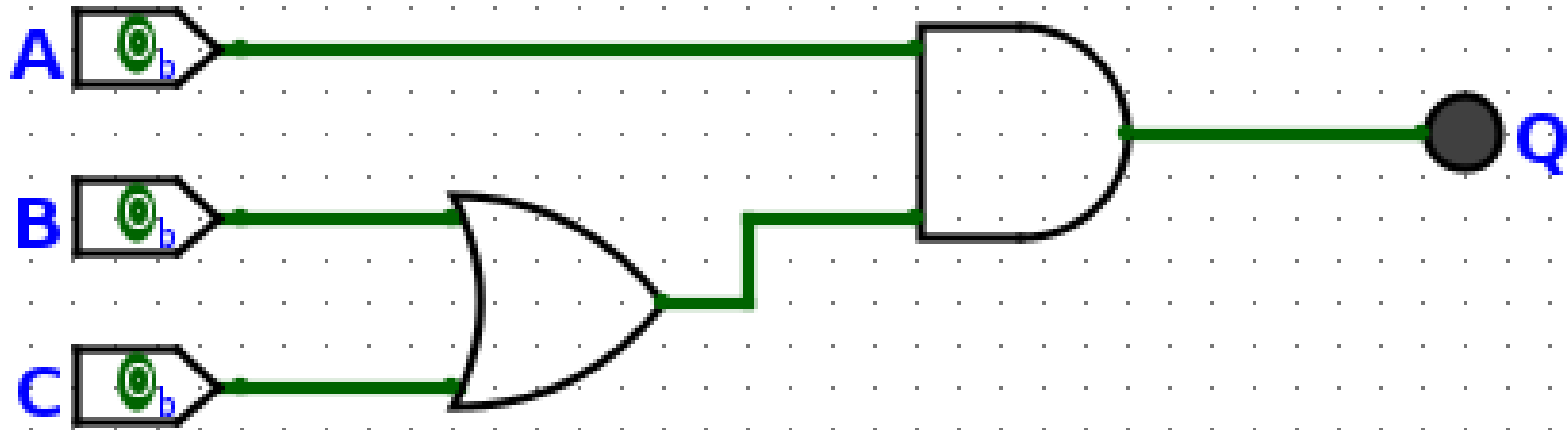
Full table Redux!

A	$R + (B \text{ OR } C)$	Q
0	0	
0	1	
0	1	
0	1	
1	0	
1	1	
1	1	
1	1	

Full table Reduced!

A	$R + (B \text{ OR } C)$	Q
0	0	0
0	1	0
1	0	0
1	1	1

Gates



Going the other way

- Often, we know the result we want, and have to work backwards

Going the other way

- Often, we know the result we want, and have to work backwards
- Addition of two numbers?

Binary addition

- The wrinkle is that Boolean gates produce a single output from individual outputs.
- We can add larger numbers by having multiple inputs, but even adding 2 input bits might require 2 output bits!

Binary addition

- $1 + 1 = 10$!
- That's OK! We just have multiple output states (like “R”) and treat the individually.

Binary addition

- Let's invent a notation for this.
- We want a result bit (state) for every column in the result.
- The columns are already “mapped” to powers, so let's use the powers. The “1s” column is “ 2^0 ” so let's call the result there R_0 .
- Then moving left we have R_1 , etc

$$0 + 0 = 00$$

A	B	R ₁	R ₀
0	0	0	0
0	1		
1	0		
1	1		

$$0 + 1 = 01$$

A	B	R ₁	R ₀
0	0	0	0
0	1	0	1
1	0		
1	1		

$$1 + 0 = 01$$

A	B	R ₁	R ₀
0	0	0	0
0	1	0	1
1	0	0	1
1	1		

$$1 + 1 = 10$$

A	B	R ₁	R ₀
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

Final truth table

A	B	R_1	R_0
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

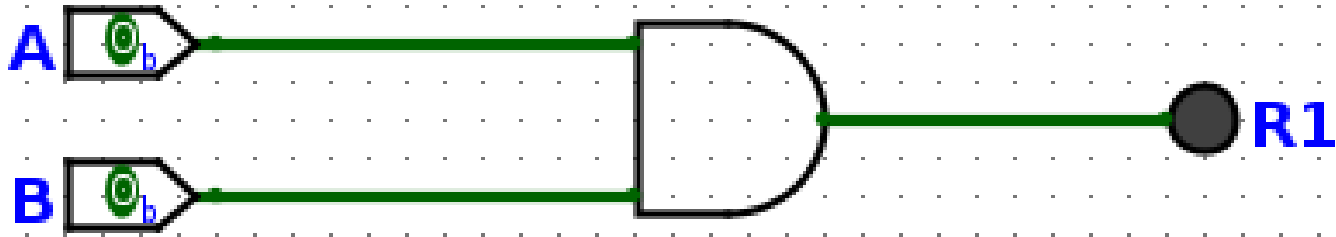
A AND B

A	B	R_1
0	0	0
0	1	0
1	0	0
1	1	1

A XOR B

A	B	R_0
0	0	0
0	1	1
1	0	1
1	1	0

If you build it....



All of it...



It will add!

